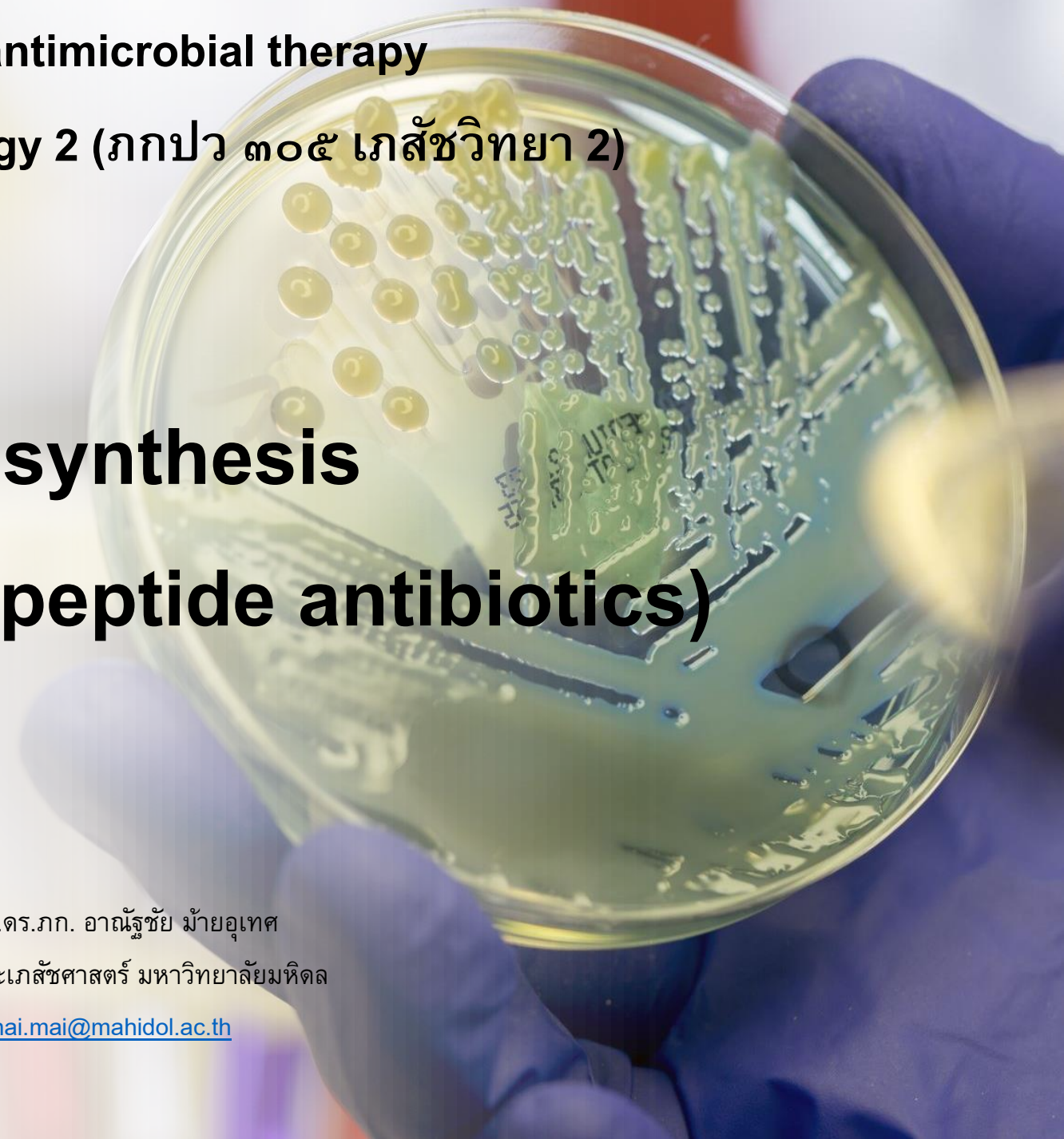


Drugs affecting cell wall synthesis (Glycopeptide/Lipoglycopeptide antibiotics)

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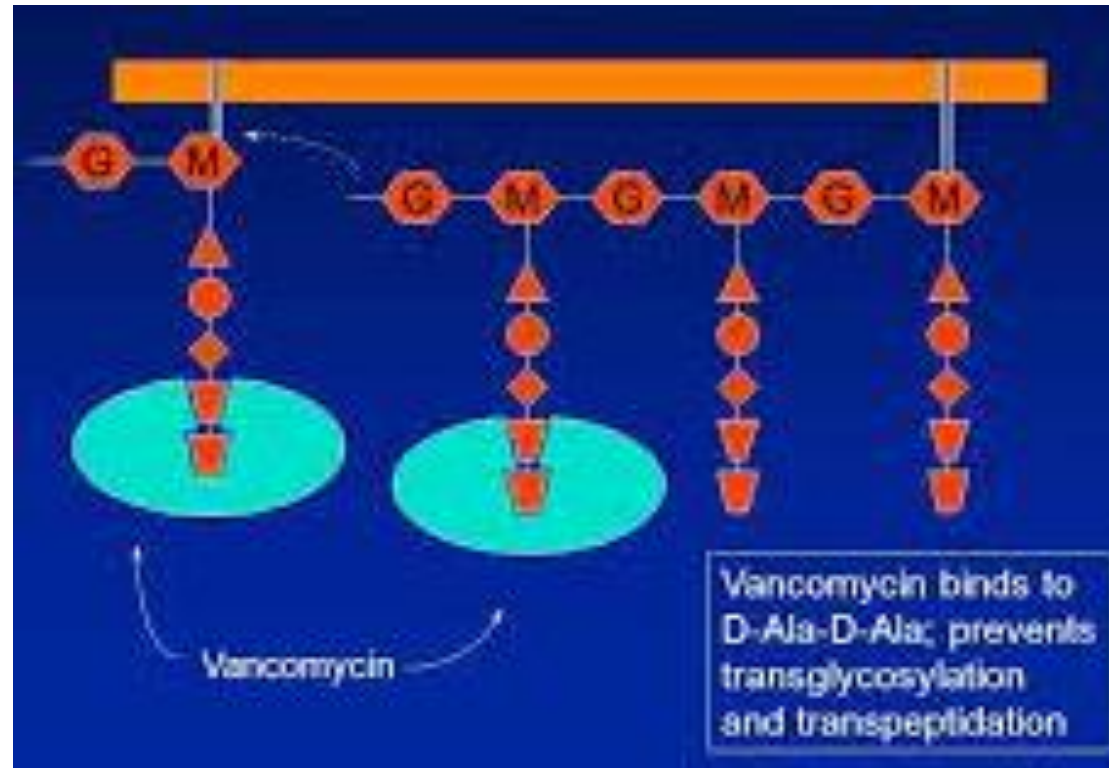
Email: arnatchai.mai@mahidol.ac.th



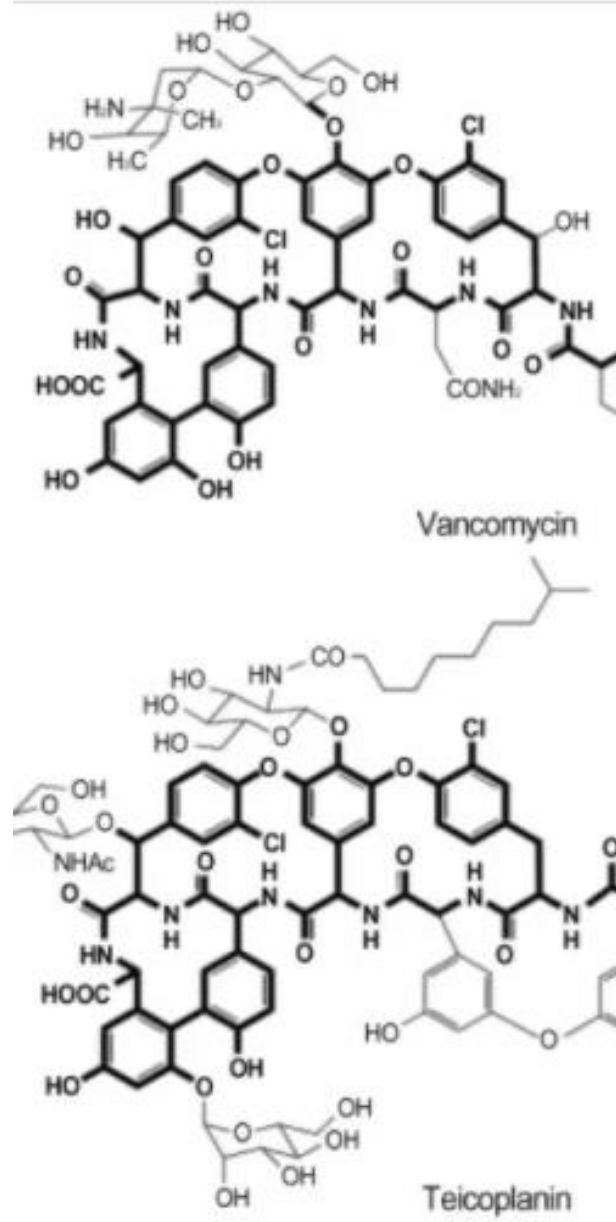


Glycopeptides: Mechanism of Action

- Inhibits cell wall synthesis by **binding to the d-Ala-d-Ala** terminus of transglycosylation and cross-linking peptidoglycan
 → prevent further elongation of peptidoglycan and cross-linking → **cell lysis**



Target of drug action



Glycopeptide antibiotics (vancomycin, teicoplanin)

two sugars moieties + an aglycone moiety

The aglycone moiety: pharmacologic activity of the molecule

The sugars moiety: hydrophilicity and its propensity to form dimers

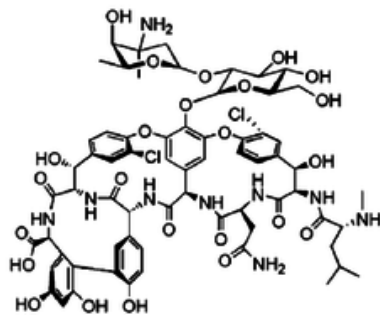
Lipoglycopeptides (dalbavancin, oritavancin and telavancin)

semisynthetic derivatives :

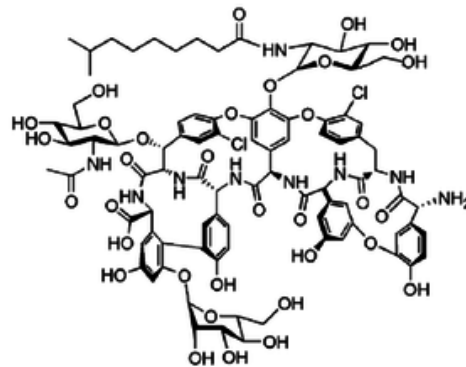
- addition of a **hydrophobic moiety**,
- additional properties to the drugs (long half-life)

Glycopeptide Antibiotics

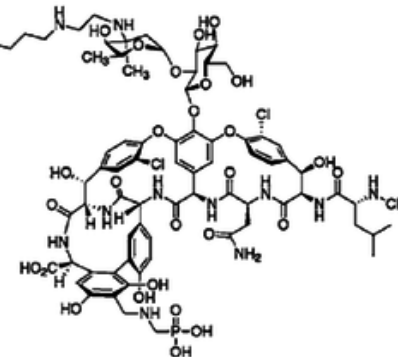
- Bactericidal
- Complex tricyclic glycopeptide, MW $\approx$ 1500 Da
- Extensively purified production to decreased adverse effects
- Today, Clinica luse increasing due to emergence of resistant bacteria (MRSA)



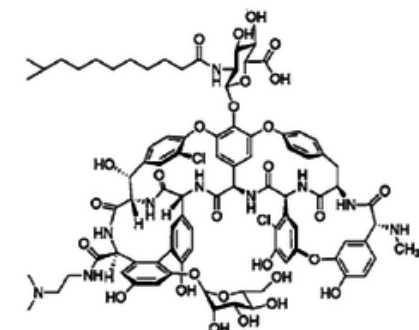
Vancomycin



Teicoplanin



Telavancin



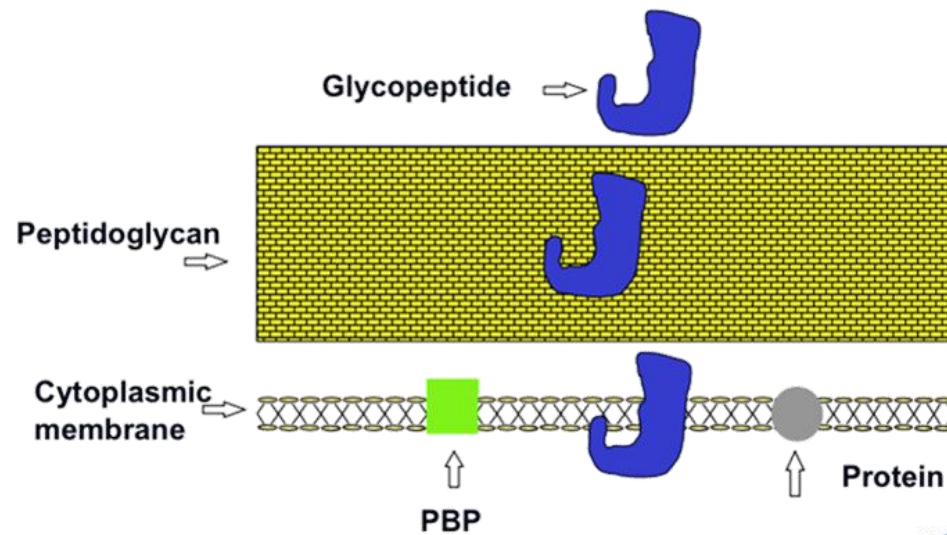
Dalbavancin

Glycopeptide : Clinical use

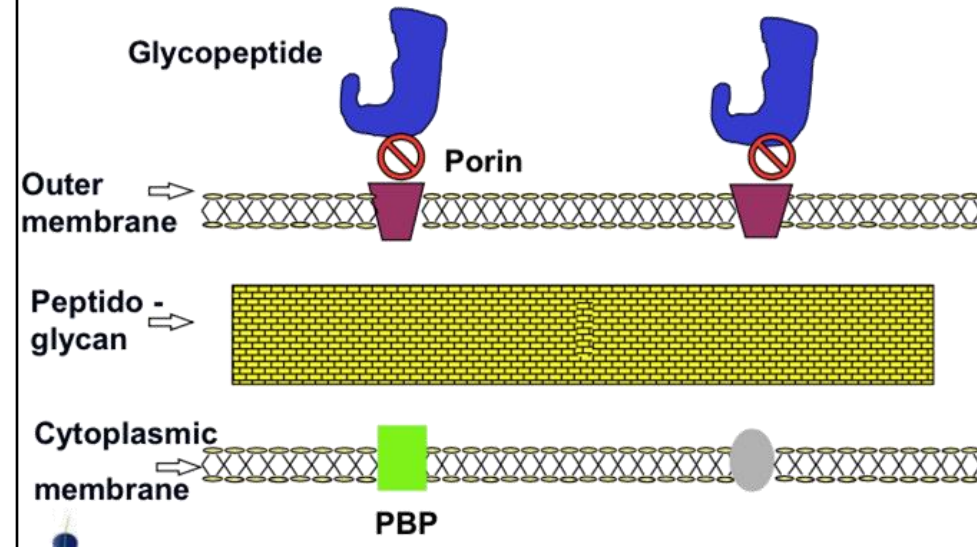
Drug	Clinical use or target pathogens	Recommended dose	ADRs
Vancomycin (IV)	- MSSA, MRSA - Enterococci (except VRE) - L. monocytogenes - Anaerobic cocci	25-30 mg/kg loading, 15-20 mg/kg q 12 h	- Redman syndrome - Nephrotoxicity

Glycopeptides: Spectrum of Action

GRAM-POSITIVE



GRAM-NEGATIVE



No activity against Gram (-) aerobes & anaerobes

Glycopeptides: Spectrum of Action

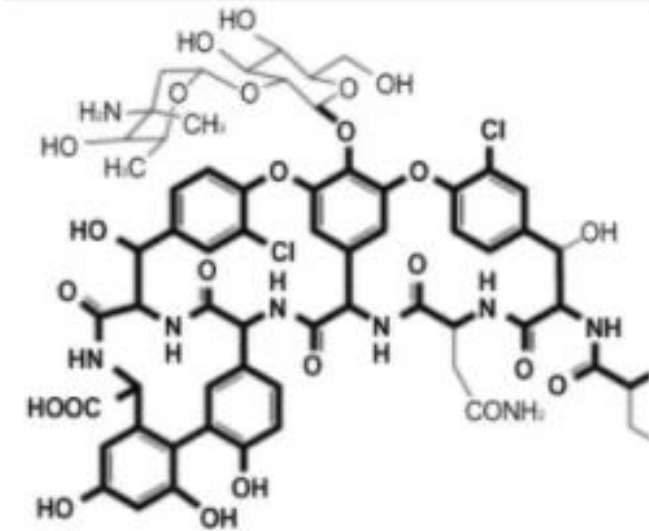
Gram (+) bacteria

- Methicillin-Susceptible & Methicillin-Resistant *S. aureus* and coagulase-negative staphylococci
- *Streptococcus pneumoniae* (including PRSP), Viridans streptococcus, Group streptococcus
- *Enterococcus* sp.
- *Corynebacterium*, *Bacillus*, *Listeria*, *Actinomyces*
- *Clostridium* sp. (including *C. difficile*), *Peptococcus*, *Peptostreptococcus*

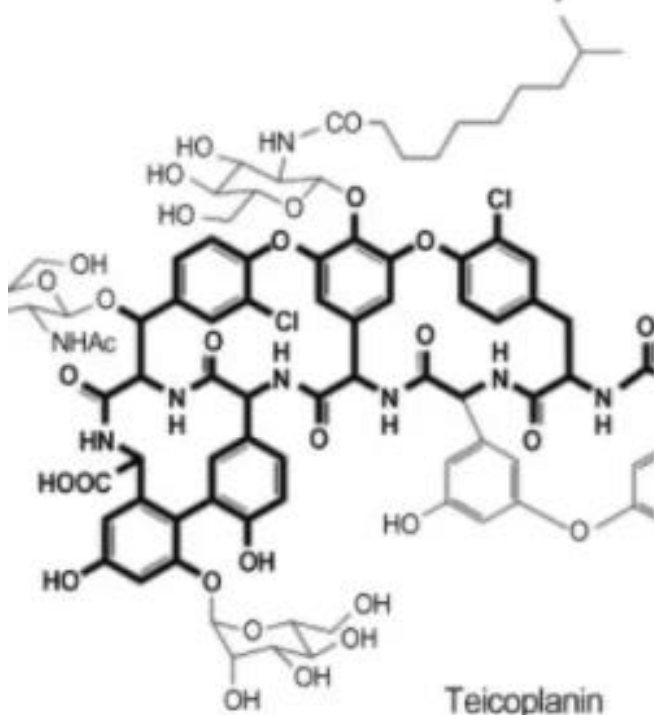
No activity against Gram (-) aerobes & anaerobes

Vancomycin: Adverse Effects

- **Red-Man Syndrome:** flushing, pruritus, erythematous rash
 - Related to **Rate of IV infusion**; should be over 60 min
 - Resolves spontaneously after discontinuation
 - May pretreat with **Antihistamines** in some cases
- **Nephrotoxicity and Ototoxicity**
 - Rare with monotherapy, more common when use with other nephro-/oto-toxins or renal impairment
- **Thrombophlebitis**



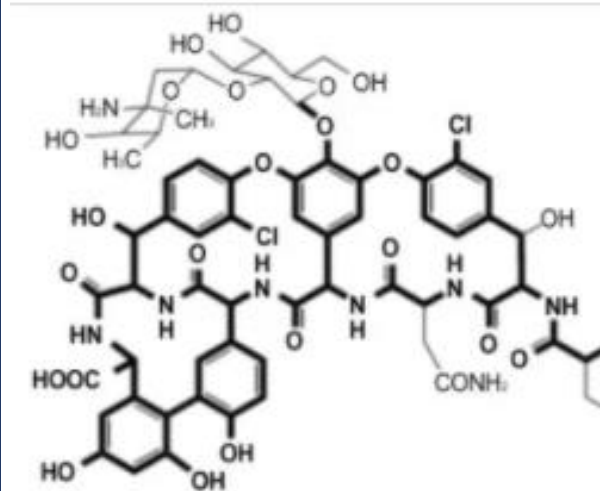
Vancomycin



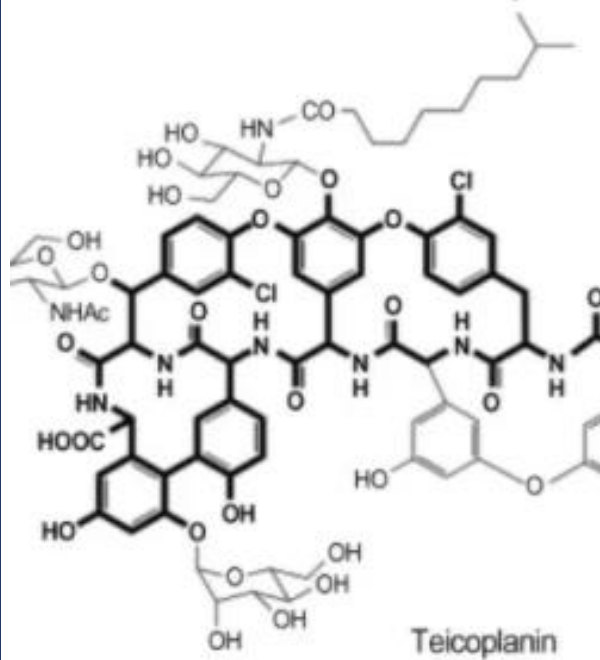
Teicoplanin

Vancomycin

- Bactericidal for **Gram+** (0.5-10 ug/ml)
- Hydrophilic drug
- *Staphylococci spp.* Including producing beta-lactamase, resistance to nafcillin and methicillin (2 ug/ml)
- Action only when bacteria actively dividing (slower than penicillin)
- Synergistic with gentamicin or streptomycin for killing *Enterococcus faecium*, *Enterococcus faecalis*



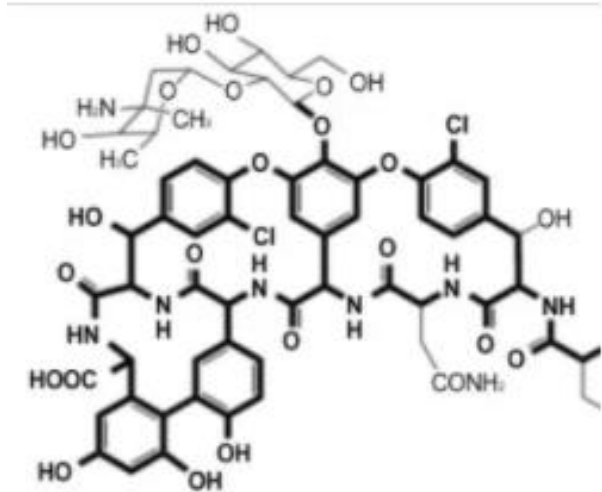
Vancomycin



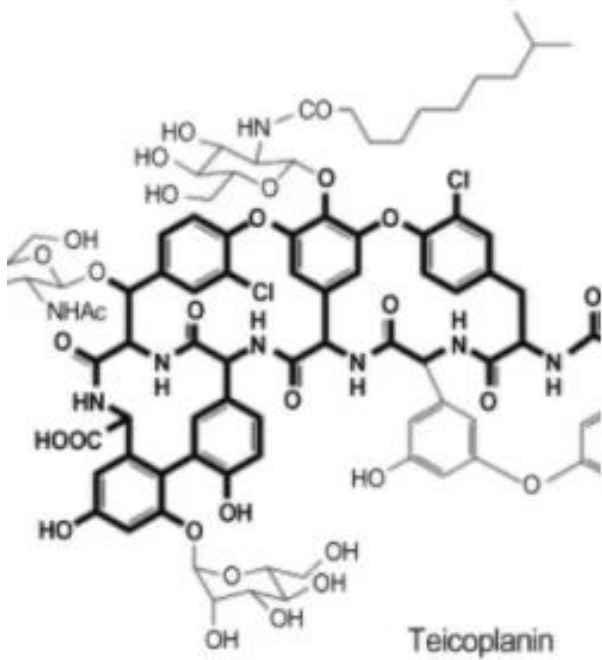
Teicoplanin

Vancomycin

- Poorly absorbed from GI
 - Oral only for treat *C. difficile*
- **1-h** intravenous infusion of 1 g produces blood levels 15-30 ug/ml
 - Prevent red man syndrome (RMS)
 - Flushing, erythema, pruritus, hypotension, myalgia
 - Stop infusion/ give anti-histamine
- Drug is widely distributed in body
 - Cerebrospinal fluid = 7-30% of serum concentration (if meningeal inflammation)



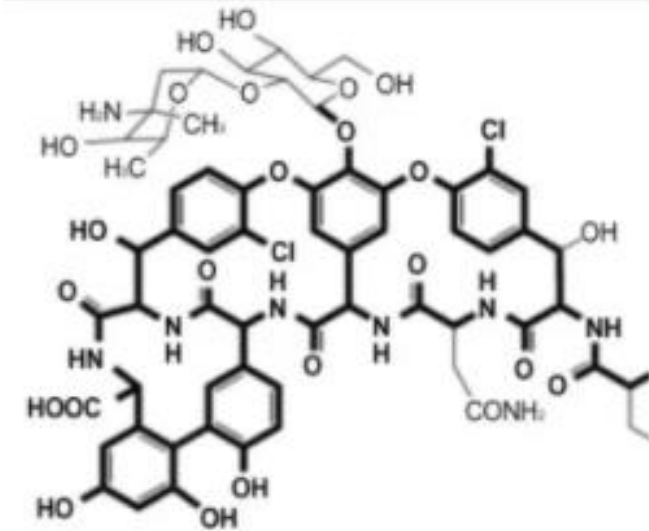
Vancomycin



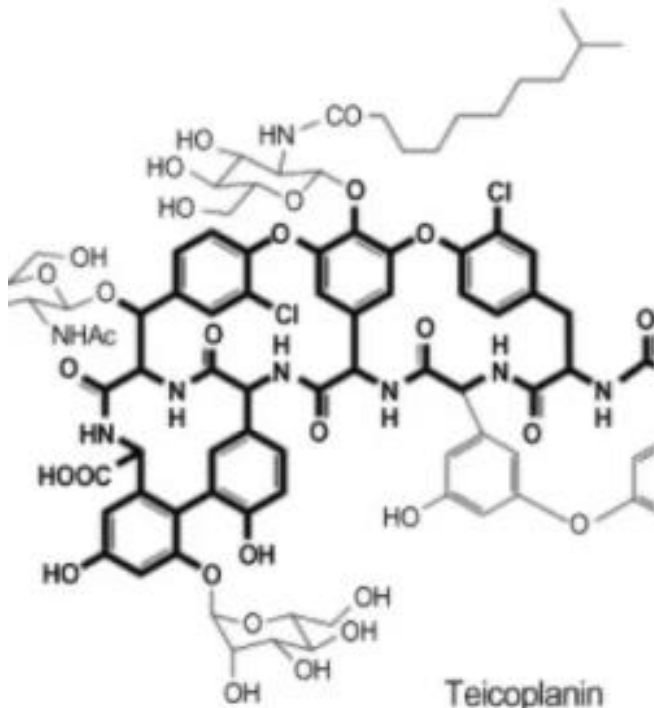
Teicoplanin

Vancomycin

- 90% excreted by glomerular filtration
- Half-life in serum about 6 h
 - In patient with insufficiently function of kidney: half-life is 6-10 days
 - Hemodialysis can remove 50% of drug
- Ototoxic can be minimized by maintaining peak serum concentration below 60 ug/ml



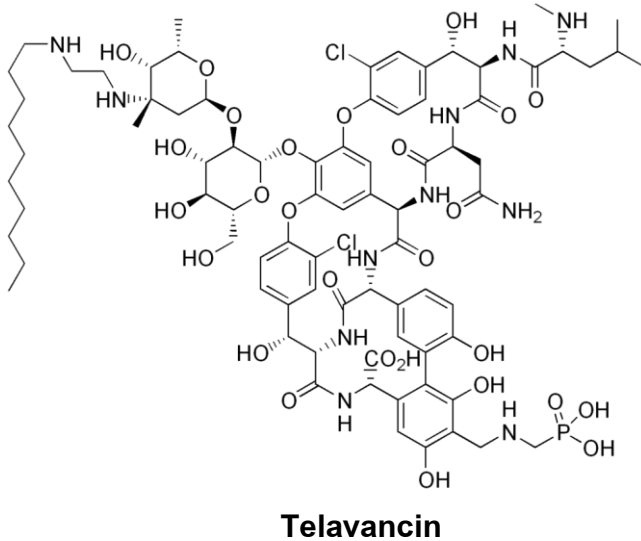
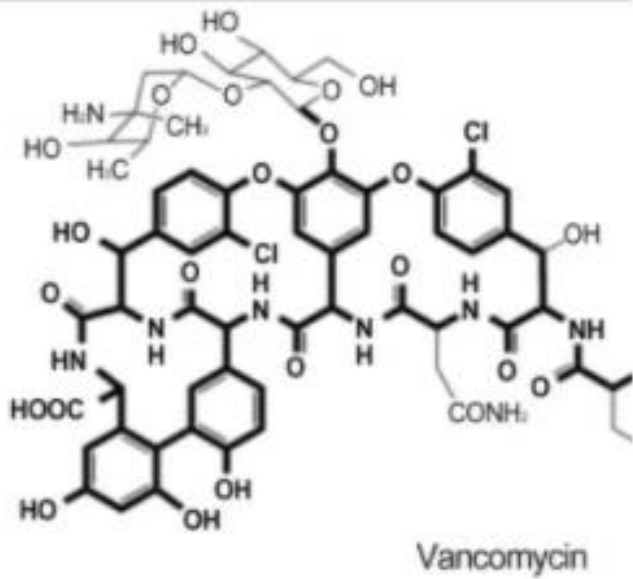
Vancomycin



Teicoplanin

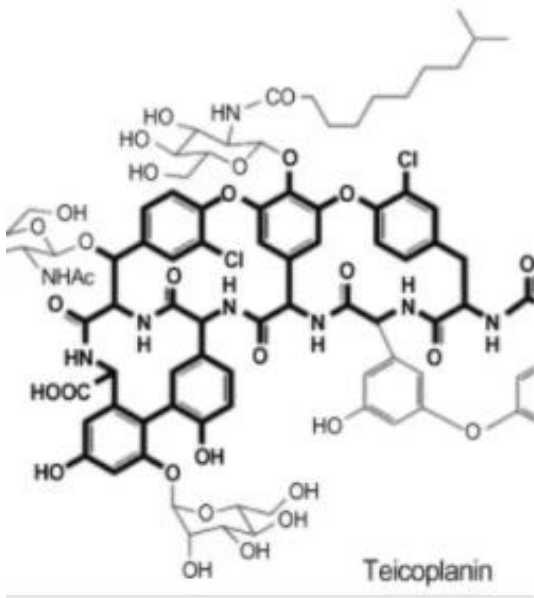
Teicoplanin

- Bacterial spectrum is similar to vancomycin
- Can be given both IM as well as IV
- Long half-life (45-70h)
 - Once daily dosing



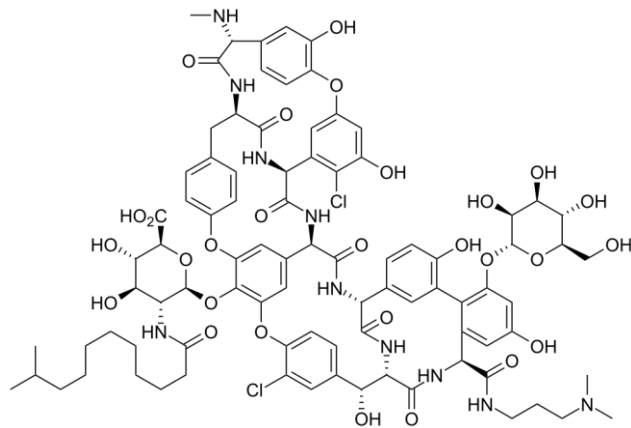
Telavancin

- Semisynthetic derived from vancomycin
- IV dosing with **half-life 8 h**
- Once daily dosing
- For complicated skin and soft tissue infection



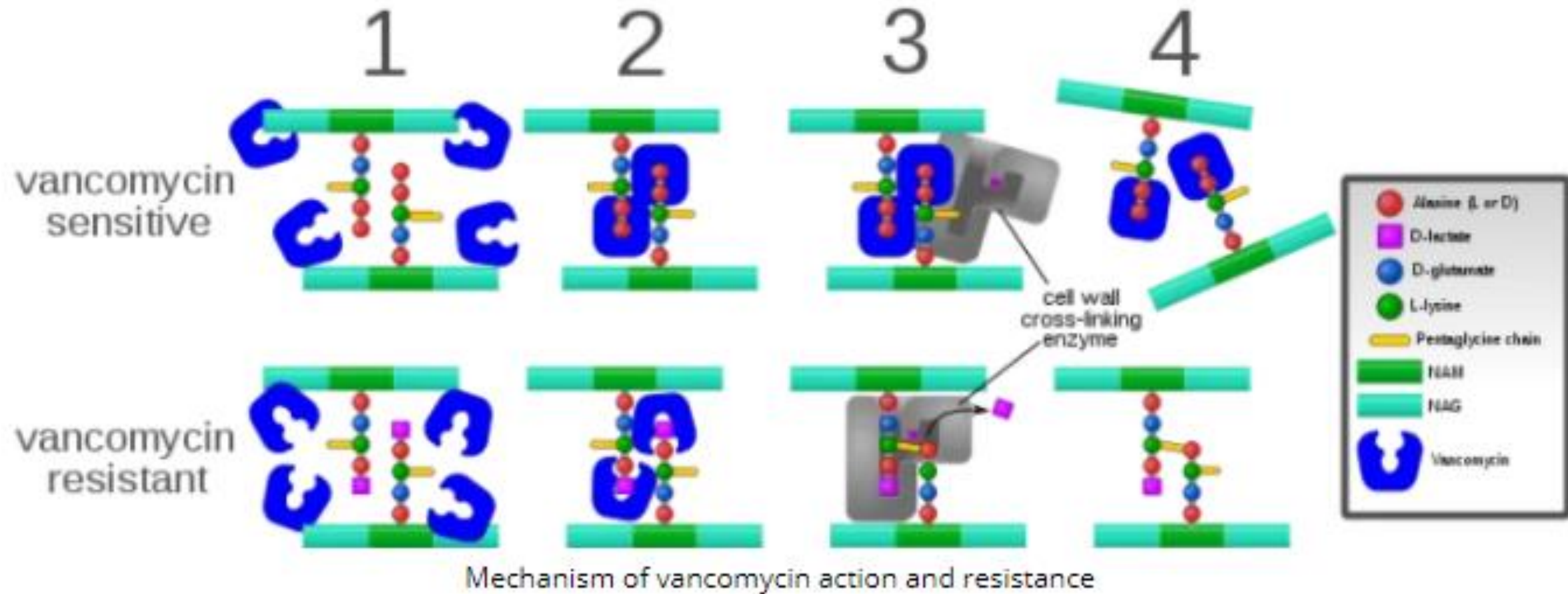
Dalbavancin

- Semisynthetic derived from Teicoplanin
- IV dosing with **extremely long half-life** about 6-11 days
 - **Once weekly** dosing
- For complicated skin and soft tissue infection



Dalbavancin

Glycopeptide resistance



Vancomycin resistant bacteria: **D-alanine to D-lactate**

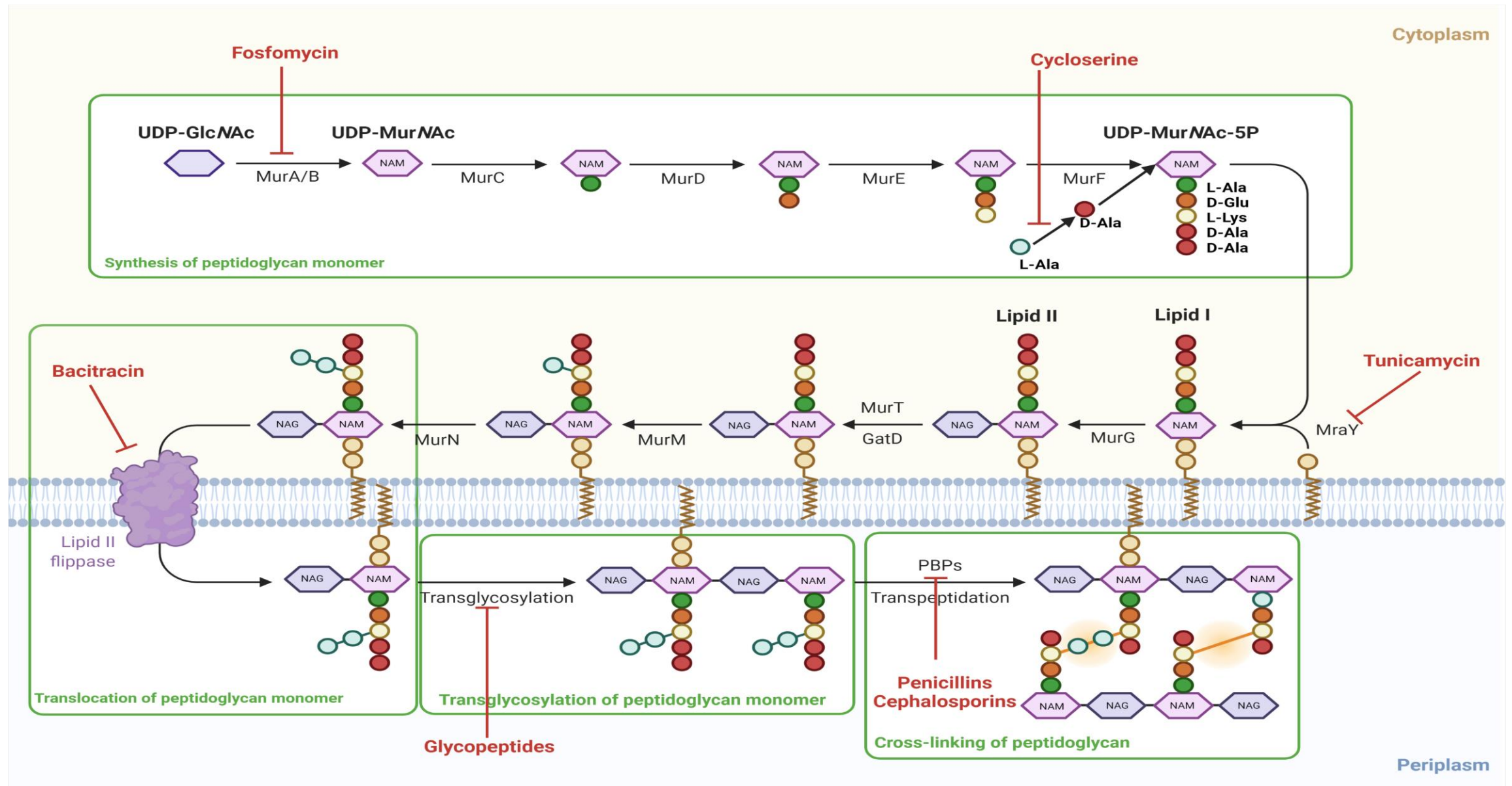
Loss of hydrogen bond and reduce affinity of drug

Vancomycin-resistant enterococcus (VRE)

- Six resistance patterns (designated “VanA” through “VanE,” and “VanG”)
- Multiple genes are responsible for these different phenotypes
 - The target for vancomycin is the terminal **d-alanyl-d-alanine** of the growing peptidoglycan chain
 - In vancomycin-resistant organisms
 - **d-alanyl-d-lactate** (VanA, VanB, and VanD)
 - **d-alanyl-d-serine** (VanC, VanE, and VanG).
 - These altered targets have **reduced affinity for vancomycin**, resulting in much higher MICs.

Isolates	Minimal Inhibitory Concentration (MIC)
Vancomycin-susceptible <i>S.aureus</i>	0.5-2 µg/mL
Vancomycin-intermediate <i>S.aureus</i> (VISA)	4-8 µg/mL
Vancomycin-resistant <i>S.aureus</i> (VRSA)	≥16 µg/mL

Target of drug action



- **Polypeptide**
 - Interfere dephosphorylation of membrane carrier molecules
 - C55-isoprenyl pyrophosphate
 - Bactroprenol pyrophosphate
- **Disrupt cell wall synthesis of Gram positive**
 - *Staphylococcus aureus*: 0.03-700 ug/ml
 - *Staphylococcus epidermidis*: 0.25-16 ug/ml
 - *Streptococcus pyogenes*: 0.5-16 ug/ml
- **Ointment**: combination with polymyxin B and neomycin

